

535510 RL Team Project Proposal

DMRL: Reinforcement Learning for Inverse Problems with Diffusion Models

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1 Project Overview

Track: Algorithm + Application

Abstract and Project Goal

Inverse problems aim to recover an unknown signal x from noisy measurements $y = A(x) + n$, where A is a (possibly nonlinear) forward model and n denotes measurement noise. Recent diffusion models have achieved strong performance for such tasks by leveraging powerful learned priors. However, existing diffusion-based reconstruction methods typically enforce measurement consistency through gradient-based updates with manually tuned guidance strengths during the diffusion process.

In this project, we propose formulating diffusion-based inverse problem reconstruction as a sequential decision-making problem and applying reinforcement learning (RL) to learn an adaptive reconstruction policy. The RL agent dynamically controls the reconstruction trajectory during diffusion sampling by adjusting the strength of measurement guidance at each step. We aim to evaluate the proposed approach on representative inverse problem tasks and compare it with existing diffusion-based reconstruction methods.

TL;DR: We use reinforcement learning to learn an adaptive measurement-guidance policy for diffusion-based inverse problems, replacing manually tuned reconstruction schedules.

1.1 Motivation

- Why is the problem interesting? Inverse problems are fundamental in many imaging and signal processing applications where the goal is to recover an unknown signal x from measurements y . Examples include image inpainting, super-resolution, and deblurring. Recently, diffusion-based generative models have demonstrated strong performance in solving such problems by leveraging powerful learned priors. By combining diffusion sampling with measurement-consistency updates, these methods can reconstruct signals that both match the measurements and lie on the natural data manifold.

Despite their success, diffusion-based inverse problem methods typically rely on manually designed update rules to balance generative prior guidance and measurement consistency during reconstruction. **Designing these schedules often requires task-specific heuristics and careful hyperparameter tuning.**

- Critical challenges.
 - **Challenge 1: Balancing generative prior and measurement consistency.** Diffusion-based inverse problem methods typically combine a denoising update from the diffusion model with a measurement correction step. Determining how strongly each component should influence the reconstruction is difficult and often requires manually tuned parameters.
 - **Challenge 2: Adaptive control of the reconstruction trajectory.** The optimal update strategy may vary across diffusion timesteps and depend on the current reconstruction quality. However, most existing methods rely on fixed or heuristic schedules that do not adapt to the current state of the reconstruction process.
 - **Challenge 3: Sequential decision nature of reconstruction.** Diffusion-based reconstruction proceeds through a sequence of iterative updates from x_T to x_0 . Each step affects future reconstruction quality, making the problem naturally sequential and potentially suitable for reinforcement learning.

- Why does the problem remain open? Most existing approaches treat the reconstruction process as a fixed optimization procedure with manually designed update rules. For example, Diffusion Posterior Sampling (DPS) [1] integrates measurement gradients into the reverse diffusion process using a predefined guidance strength. While effective, such methods rely on hand-designed schedules that may not generalize well across different tasks or measurement settings. Learning an adaptive reconstruction policy that dynamically controls the update process remains an open research direction.
- Relevant Prior Work. Recent work has explored diffusion models for inverse problems, including Diffusion Posterior Sampling (DPS) [1], RePaint [2] for image inpainting, and other plug-and-play diffusion reconstruction approaches [3]. These methods combine generative priors with measurement guidance but typically rely on manually chosen update rules.

State-of-the-art

Among existing approaches, Diffusion Posterior Sampling (DPS) [1] is widely regarded as a strong baseline for solving inverse problems using diffusion models, as it integrates measurement guidance directly into the reverse diffusion process while preserving the generative prior.

2 Problem Formulation

We consider the inverse problem of recovering an unknown signal $x \in \mathbb{R}^d$ from noisy measurements

$$y = A(x) + n,$$

where $A(\cdot)$ denotes the forward measurement operator and n represents measurement noise. The goal is to reconstruct x given the observation y . Recent approaches leverage diffusion generative models as powerful priors for solving such problems. In diffusion-based reconstruction methods, the reverse diffusion process generates samples through a sequence of iterative updates

$$x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0,$$

starting from an initial Gaussian noise sample $x_T \sim \mathcal{N}(0, I)$. At each timestep t , the diffusion model predicts a denoising update that moves the sample toward the data manifold.

To incorporate measurement consistency, many diffusion-based inverse problem methods introduce a measurement correction step during sampling. For example, a typical update can be written as

$$x_{t-1} = x'_{t-1} - \gamma_t \nabla_x \|y - A(\hat{x}_0)\|^2,$$

where x'_{t-1} denotes the reverse diffusion update, $\hat{x}_0$ is the predicted clean signal, and γ_t controls the strength of measurement guidance.

In existing methods, the guidance parameter γ_t is typically chosen using a fixed or heuristic schedule. In this work, we instead view the reconstruction process as a sequential decision-making problem and aim to learn an adaptive policy that dynamically controls the reconstruction trajectory during diffusion sampling.

3 Empirical Evaluation

We evaluate the effectiveness of the proposed RL-based reconstruction policy by comparing it with existing diffusion-based inverse problem solvers. The evaluation focuses on reconstruction quality and measurement consistency across several representative inverse problem tasks.

3.1 Performance Metrics

We adopt the following metrics to evaluate reconstruction performance:

- **PSNR** (Peak Signal-to-Noise Ratio). PSNR measures the pixel-level reconstruction accuracy between the reconstructed signal and the ground truth image. It is a standard metric used in image restoration tasks.
- **SSIM** (Structural Similarity Index). SSIM evaluates the perceptual similarity between two images by comparing structural information, providing a complementary measure to PSNR.
- **LPIPS** (Learned Perceptual Image Patch Similarity). LPIPS measures perceptual similarity using deep neural network features and is widely used to evaluate visual quality in generative modeling and image reconstruction tasks.
- **Measurement Consistency Error**. We measure the residual error $\|y - A(x)\|$ to evaluate how well the reconstructed signal satisfies the measurement constraint.
- **Inference Efficiency**. We also report the computational cost of reconstruction, including the number of diffusion steps and total inference time.

3.2 Baseline Methods

We compare our method with the following diffusion-based inverse problem approaches:

- **Diffusion Posterior Sampling (DPS)** [1]. A diffusion-based reconstruction method that incorporates measurement gradients during the reverse diffusion process.
- **RePaint** [2]. A diffusion-based inpainting algorithm that alternates between denoising and resampling steps to enforce measurement constraints.
- **DMPlug** [3]. A diffusion-based plug-and-play method that integrates measurement information during the diffusion sampling process to improve reconstruction quality for inverse problems.

3.3 Benchmark Tasks or Datasets

We evaluate the proposed method on several representative inverse problem tasks in image restoration.

Image Inpainting. Random regions of the input image are masked and the model must reconstruct the missing pixels. This task evaluates the ability to incorporate measurement constraints while maintaining realistic image structure.

Image Super-Resolution. The model reconstructs a high-resolution image from a low-resolution observation produced by a downsampling operator.

Image Deblurring. The measurement operator applies a convolutional blur kernel to the original image. The reconstruction task is to recover the sharp image from the blurred observation.

For all tasks, we plan to conduct experiments on standard 256x256 image datasets such as CelebA and FFHQ, which are commonly used benchmarks in diffusion-based generative modeling.

3.4 Baseline Evaluation and Toy Example

Before training a reinforcement learning policy, we first evaluate whether adaptive measurement guidance is useful for diffusion-based inverse problem reconstruction. The goal of this baseline evaluation is to verify that different temporal designs of the guidance parameter η_t can significantly affect reconstruction quality.

We consider the inverse problem of recovering a clean image x from a noisy measurement

$$y = A(x) + n, \tag{1}$$

where $A(\cdot)$ denotes the forward measurement operator and n denotes measurement noise. During diffusion sampling, measurement consistency is enforced by applying a correction step controlled by the guidance strength η_t . Instead of using a fixed guidance strength, we study whether changing η_t over diffusion timesteps can improve the balance between measurement consistency and perceptual quality.

We compare the following guidance schedules:

- **Paper baseline η :** This setting follows the original baseline guidance schedule used in the diffusion posterior sampling framework. It serves as the reference method.
- **Constantly large η :** This setting uses a consistently large measurement guidance strength throughout the reverse diffusion process. It encourages stronger measurement consistency, but it may overfit the degraded observation or measurement noise.
- **Constantly small η :** This setting uses a consistently small guidance strength. It allows the diffusion prior to dominate the reconstruction process, but the reconstructed image may fail to sufficiently satisfy the measurement constraint.
- **Hand-designed adaptive η_t :** This schedule starts with a small value in the early reverse diffusion steps, increases the guidance strength in the middle steps, and decreases it near the final steps. The purpose is to enforce measurement consistency when the image structure becomes meaningful while avoiding overfitting to the measurement noise at the end.

The hand-designed adaptive schedule follows the intuition that different diffusion timesteps play different roles. In the early stage, the sample is still highly noisy, so applying a strong measurement correction may be unstable. In the middle stage, the image structure becomes clearer, so stronger measurement guidance can help align the reconstruction with the observation. In the late stage, overly strong guidance may force the reconstruction to match noise or blur artifacts in the measurement, so reducing η_t helps preserve perceptual image quality.

The adaptive schedule can be summarized as

$$\eta_t = \begin{cases} \text{small,} & \text{early reverse diffusion steps,} \\ \text{large,} & \text{middle reverse diffusion steps,} \\ \text{small,} & \text{late reverse diffusion steps.} \end{cases} \quad (2)$$

For the toy example, we conduct experiments on this FFHQ image with a motion deblurring inverse problem:



Figure 1: Image 00015 in the FFHQ dataset

The measurement operator is a motion blur kernel with kernel size 61 and intensity 0.5. Gaussian noise with standard deviation $\sigma = 0.05$ is added to the measurement. For the adaptive guidance setup, we use

$$\eta_{\min} = 0, \quad \eta_{\max} = 5, \quad (3)$$

with 1000 diffusion timesteps.

We evaluate reconstruction quality using PSNR, SSIM and LPIPS. PSNR and SSIM measure pixel-level and structural reconstruction quality, LPIPS evaluates perceptual similarity. We run the test 10 times for each case and evaluate the mean and the standard deviation of each metric.

This toy example provides an important motivation for reinforcement learning. If the hand-designed adaptive schedule outperforms the paper baseline and both constant schedules, it suggests that the guidance strength should not remain fixed during sampling. Instead, the reconstruction process benefits from dynamically adjusting η_t over time.

However, the hand-designed adaptive schedule still requires manual tuning and task-specific intuition. Therefore, after validating the usefulness of adaptive guidance, we will implement a REINFORCE algorithm to learn the guidance policy automatically. In this formulation, the agent observes the current reconstruction state and selects the guidance strength η_t as its action. The reward encourages improvement in reconstruction quality. This allows the policy to learn when to increase or decrease measurement guidance during the diffusion sampling process, instead of relying on a manually designed schedule. This will be done after the confirmation of our current toy tasks, in the next project update.

Table 1: Quantitative comparison of different constant η values on FFHQ motion deblurring. The motion blur kernel size is 61, the blur intensity is 0.5, and the Gaussian noise level is $\sigma = 0.05$. Higher PSNR and SSIM are better, while lower LPIPS is better.

η	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
0.01	17.15 ± 0.88	0.60 ± 0.05	0.49 ± 0.04
0.1	22.86 ± 1.88	0.73 ± 0.03	0.31 ± 0.03
0.3	23.72 ± 2.48	0.67 ± 0.12	0.32 ± 0.08
0.5	24.78 ± 2.38	0.76 ± 0.04	0.24 ± 0.03
1	26.50 ± 3.28	0.80 ± 0.05	0.22 ± 0.04
5	23.24 ± 5.95	0.67 ± 0.22	0.33 ± 0.19
10	20.53 ± 6.80	0.50 ± 0.23	0.48 ± 0.19

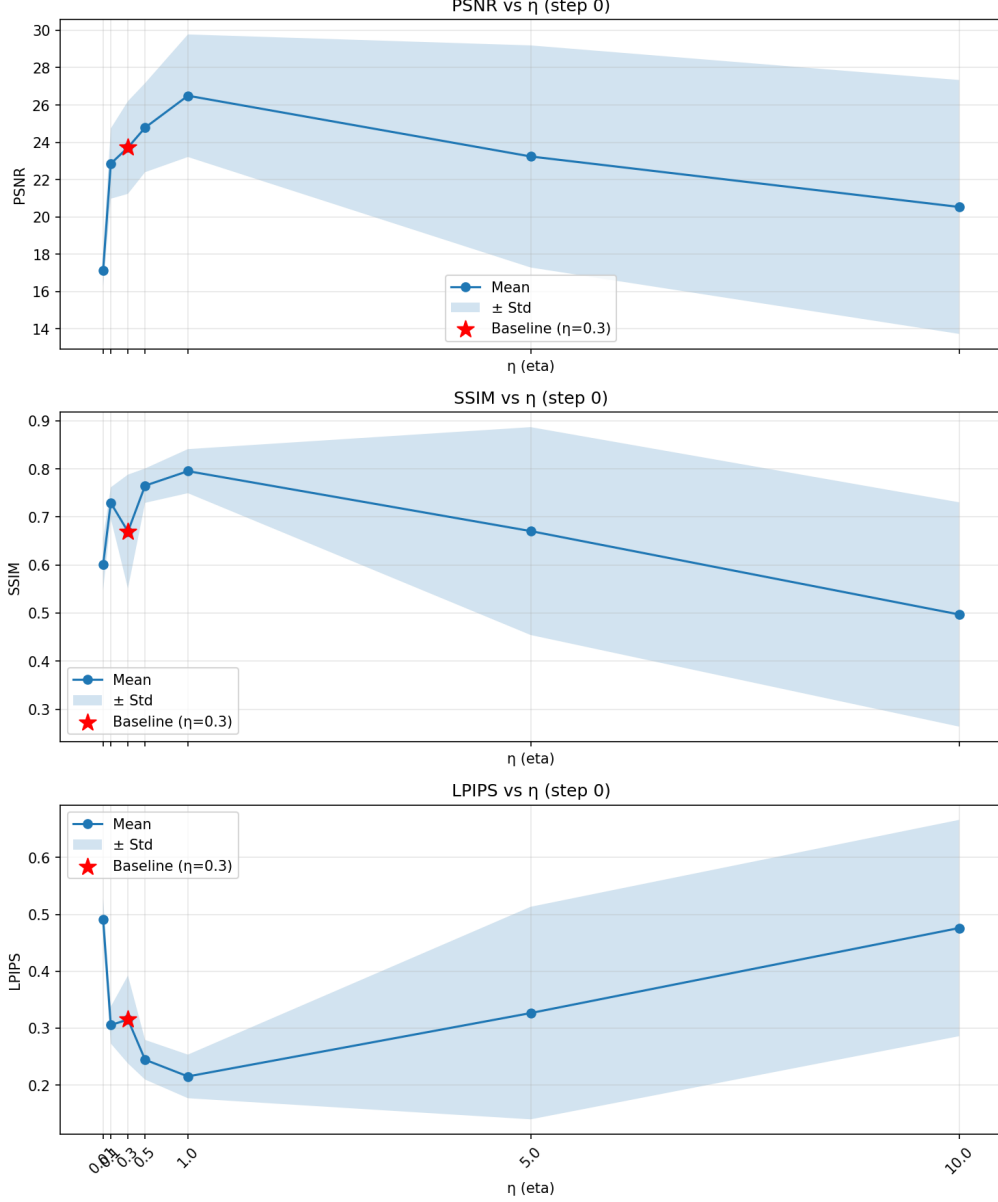


Figure 2: Visualization of metrics values under different guidance schedules. In red, the baseline

We can clearly see that the constant $\eta = 0.3$ is not the perfect value for this image.

4 Methodology

We propose to learn an adaptive reconstruction policy for diffusion-based inverse problems using reinforcement learning (RL). The reconstruction process is modeled as a Markov decision process (MDP) in which an agent controls the measurement guidance strength during diffusion sampling.

4.1 MDP Formulation

We define the MDP as a tuple $(\mathcal{S}, \mathcal{A}, P, r)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P represents the transition dynamics, and r is the reward function.

State. At timestep t , the state summarizes the current reconstruction status. We define the state as

$$s_t = \left(\frac{t}{T}, \|y - A(\hat{x}_0)\| \right),$$

t : diffusion timestep
 T : total number of time steps
 y : input blurred image
 A : blur function
 $\hat{x}_0$: projected final image
 $\|y - A(\hat{x}_0)\|$: current measurement error

Action. The agent selects an action that determines the strength of measurement guidance during the reconstruction update. Specifically, the action is defined as

$$a_t = \gamma_t,$$

where γ_t controls the magnitude of the measurement correction applied at timestep t .

Transition. Given state s_t and action a_t , the environment performs a reverse diffusion step followed by a measurement correction step:

$$x_{t-1} = x'_{t-1} - \eta_t \nabla_x \|y - A(\hat{x}_0)\|^2,$$

where x'_{t-1} is the standard reverse diffusion update and $\hat{x}_0$ is the predicted clean signal from the diffusion model.

Reward. The reward reflects improvement in measurement consistency during reconstruction. A simple reward function is defined as

$$r_t = w_{\text{PSNR}} \times \text{PSNR}(\hat{x}_0) - w_{\text{consistency}} \times \|y - A(\hat{x}_0)\|^2 - w_{\text{TV}} \times L_{\text{TV}}$$

which provides positive reward when the reconstruction becomes more consistent with the measurement.

TV stands for total deviation, which indicates the smoothness of the image.

4.2 Policy Learning

We parameterize the policy $\pi_\theta(a_t|s_t)$ using a neural network that outputs the guidance strength γ_t . The policy is trained using a reinforcement learning algorithm such as Proximal Policy Optimization (PPO) to maximize the expected cumulative reward over the reconstruction trajectory.

5 Results

5.1 Comparison between the baseline and our RL model

We still use here the same image for the same task motion blur, and we get the mean and the standard deviation of the metrics over 10 runs.

Table 2: Quantitative comparison of baseline and RL methods on FFHQ motion deblurring. Higher PSNR and SSIM are better, while lower LPIPS is better.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
baseline	25.29 ± 2.00	0.77 ± 0.03	0.25 ± 0.04
rl	29.64 ± 1.68	0.84 ± 0.02	0.16 ± 0.03

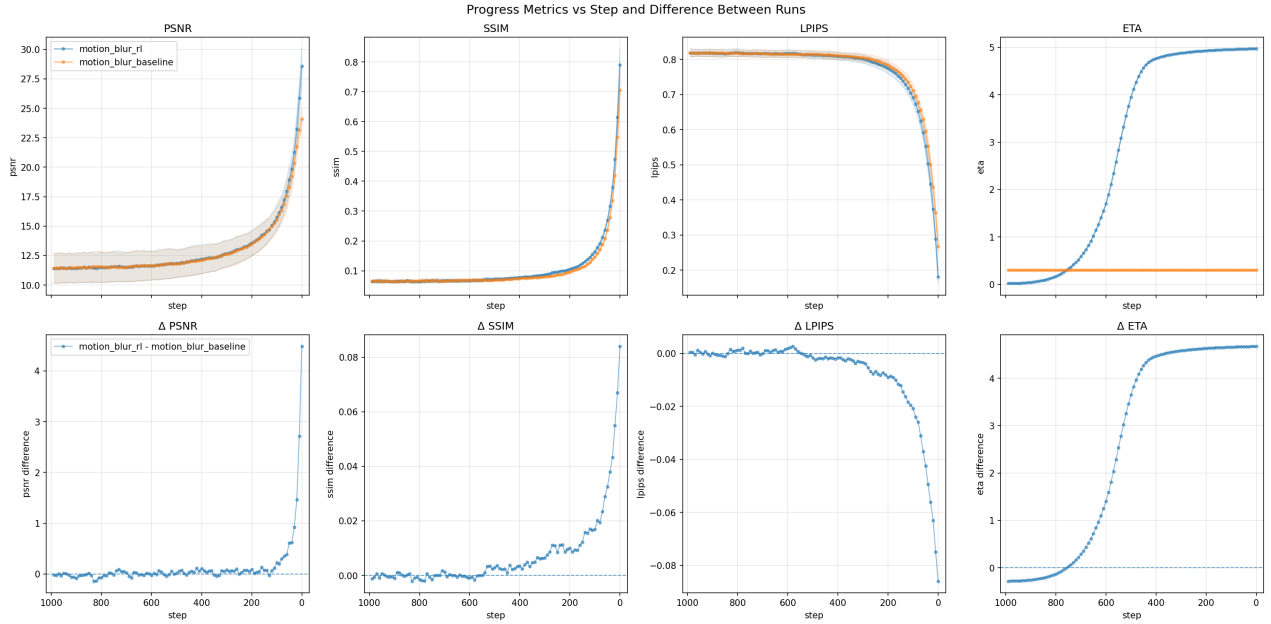


Figure 3: Visualization of metrics values under time for rl and baseline eta.

We observe that in average, the rl method has 4.45 points more for PSNR than the baseline, which is really good; it also outperform the baseline for SSIM and LPIPS.

The value of η indeed variates for each time step, as illustrated in the figure. But very surprisingly, the shape of η by time is a sigmoid function !

5.2 Comparison between the baseline and our RL model for different number of steps

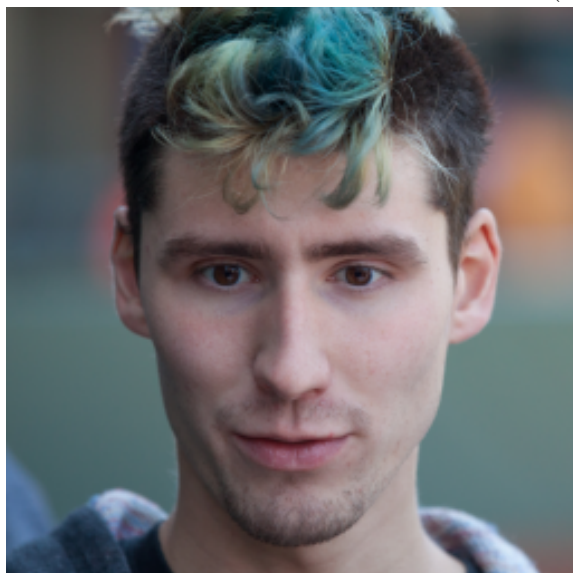
Starting from this subsection, we will use this images, and for each image run our model and the baseline DPS 10 times:



(a) Image 00003



(b) Image 00014



(c) Image 00015

Figure 4: Images in the FFHQ dataset

In this specific subsection, we will compare the metrics of our rl model and the baseline DPS with different number of total steps, and we will compare them with the motion blur task.

Table 3: Quantitative comparison of baseline and RL methods under different numbers of sampling steps on FFHQ motion deblurring. Higher PSNR and SSIM are better, while lower LPIPS is better.

Steps	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
200	baseline	9.27 ± 0.51	0.02 ± 0.00	0.90 ± 0.02
	rl	13.10 ± 1.26	0.07 ± 0.04	0.81 ± 0.04
400	baseline	12.43 ± 1.38	0.06 ± 0.02	0.83 ± 0.04
	rl	22.68 ± 1.91	0.43 ± 0.08	0.45 ± 0.03
600	baseline	19.40 ± 1.21	0.38 ± 0.07	0.50 ± 0.09
	rl	26.69 ± 1.93	0.69 ± 0.07	0.27 ± 0.03
800	baseline	23.15 ± 1.52	0.65 ± 0.09	0.30 ± 0.03
	rl	27.77 ± 1.42	0.77 ± 0.06	0.19 ± 0.02
1000	baseline	24.09 ± 0.97	0.71 ± 0.06	0.27 ± 0.02
	rl	28.58 ± 1.81	0.79 ± 0.07	0.18 ± 0.02
1200	baseline	24.46 ± 1.76	0.72 ± 0.08	0.26 ± 0.04
	rl	29.25 ± 1.36	0.80 ± 0.05	0.18 ± 0.01

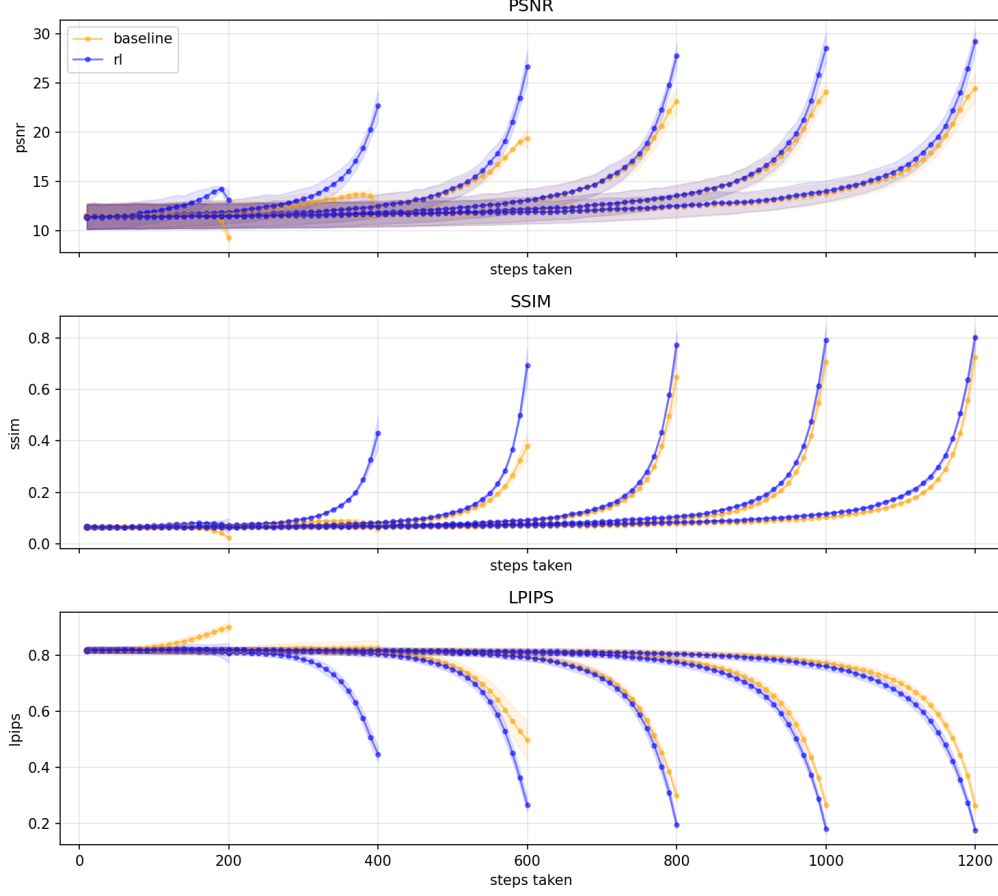


Figure 5: Visualization of metrics values with different number of steps for rl and baseline eta.

We can see that our model outperform the baseline for every number of steps, and for every metrics.

The standard version uses 1000 steps. However our rl model with only 600 has comparable metrics with the standard model with 1000 steps. Thus our model can already be used as a solution with less inference cost.

5.3 Comparison between our RL model and other hand designed methods

In this subsection, we still test with the motion blur task. We want to see if our rl model prediction for eta as a sigmoid is indeed the best solution or just a mere coincidence.

Table 4: Quantitative comparison of motion blur methods. Higher PSNR and SSIM are better, while lower LPIPS is better.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
rl	28.58 ± 2.66	0.79 ± 0.06	0.18 ± 0.04
sin	24.10 ± 3.35	0.69 ± 0.09	0.27 ± 0.07
linear	27.75 ± 2.22	0.78 ± 0.06	0.19 ± 0.04
jump	28.08 ± 1.79	0.79 ± 0.04	0.19 ± 0.03
exp	27.84 ± 2.67	0.79 ± 0.06	0.19 ± 0.04
sigmoid	28.29 ± 1.58	0.79 ± 0.05	0.19 ± 0.03

We can see that the sigmoid seems to be indeed the perfect solution.

Following the detail of each method.

I will note progress p as the progression in percentage during the inverse process (at the end $t = 0$ and $p = 1$).

$$p(t) = \frac{T - t}{T} \quad \text{with } T \text{ the total number of steps}$$

Sin:

$$\eta_{\sin}(t) = \eta_{\min} + (\eta_{\max} - \eta_{\min}) \times \sin(p(t) \times \pi)$$

$$\eta_{\min} = 0$$

$$\eta_{\max} = 5$$

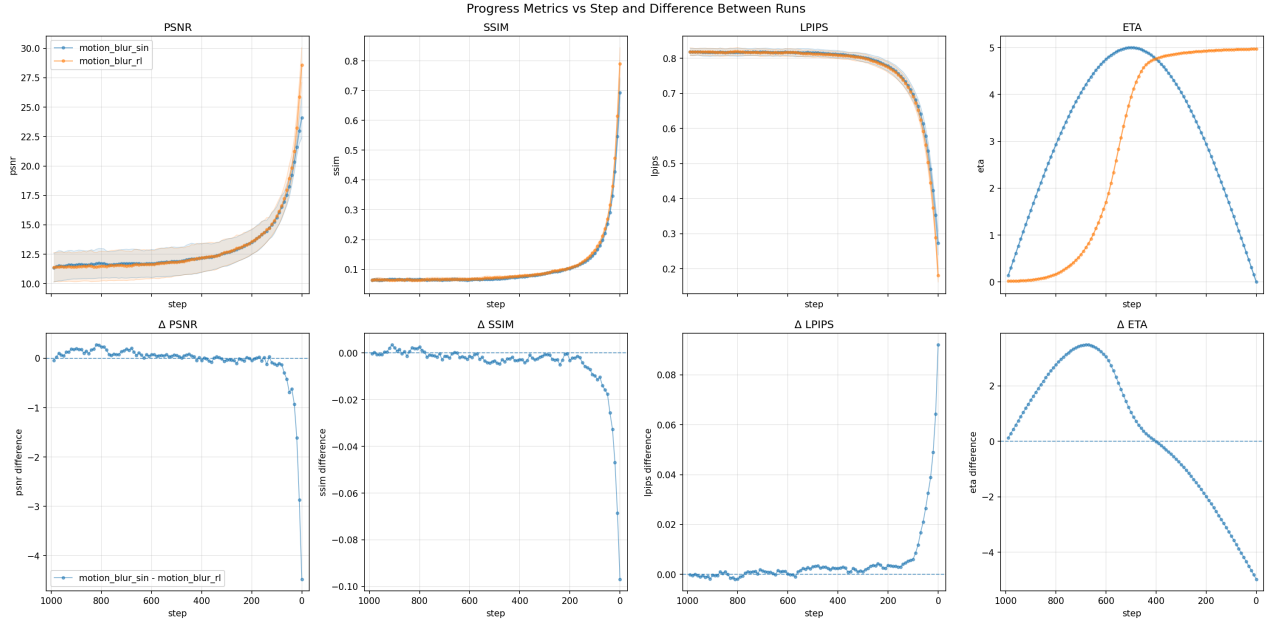


Figure 6: Comparison of sin method with our RL method.

Linear:

$$\eta(t) = \eta_{\min} + (\eta_{\max} - \eta_{\min}) \times p(t)$$

$$\eta_{\min} = 0$$

$$\eta_{\max} = 5$$

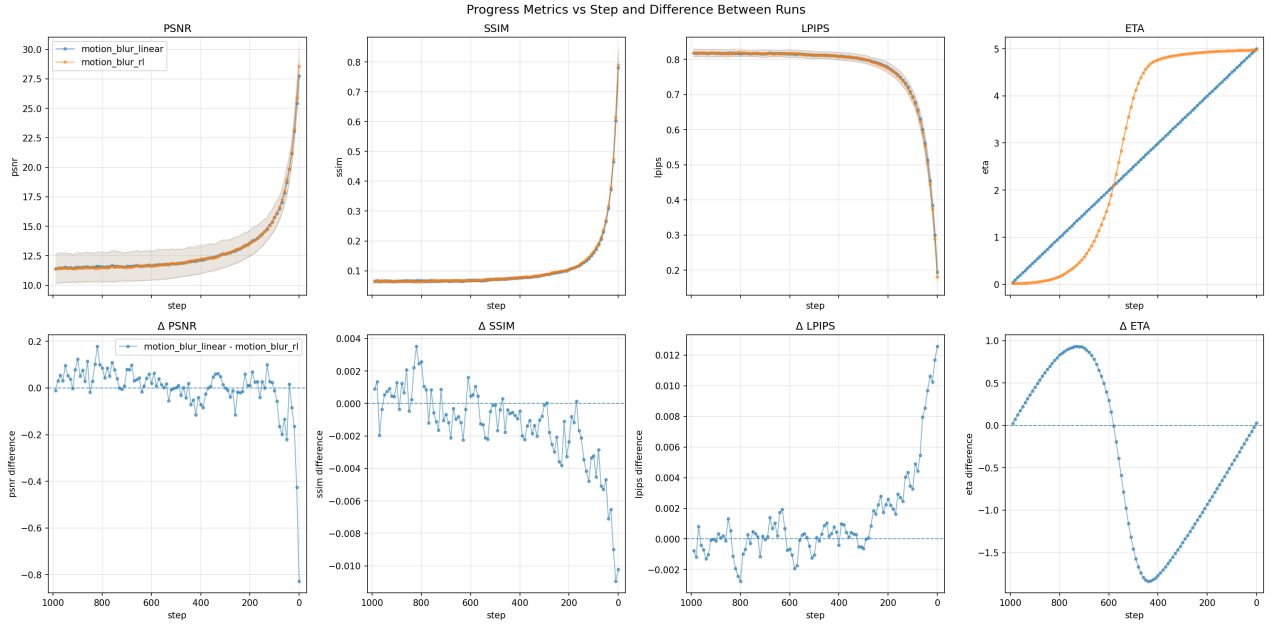


Figure 7: Comparison of linear method with our RL method.

Jump:

$$\eta_j(t) = \eta_{\min} \text{ if } p(t) < \text{mid}, \text{ else } \eta_{\max}$$

$$\eta_{\min} = 0$$

$$\eta_{\max} = 5$$

$$\text{mid} = 0.5$$

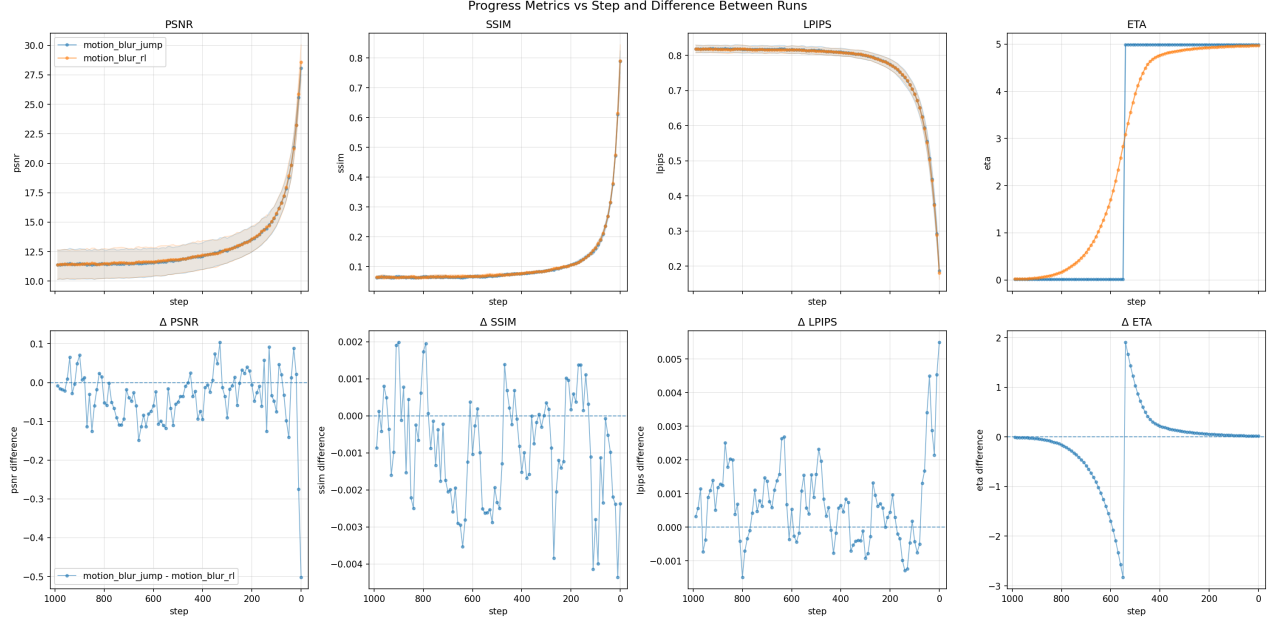


Figure 8: Comparison of jump method with our RL method.

Exp:

$$\eta_{\exp}(t) = \eta_{\min} + (\eta_{\max} - \eta_{\min}) \times \exp(\text{soft} \times (p(t) - 1))$$

$$\eta_{\min} = 0$$

$$\eta_{\max} = 5$$

$$\text{soft} = 5$$

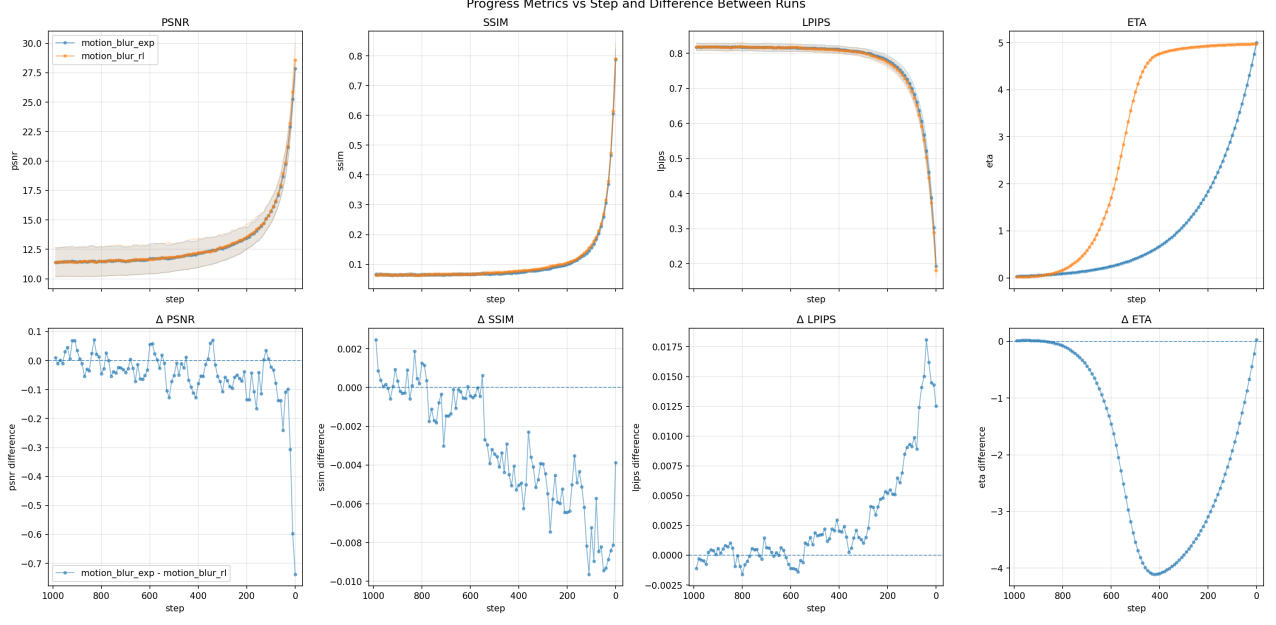


Figure 9: Comparison of exp method with our RL method.

Sigmoid:

$$\eta_{\text{sig}}(t) = \eta_{\min} + \frac{\eta_{\max} - \eta_{\min}}{1 + \exp(\text{soft} \times (\text{mid} - p(t)))}$$

$$\eta_{\min} = 0$$

$$\eta_{\max} = 5$$

$$\text{mid} = 0.45$$

$$\text{soft} = 15$$

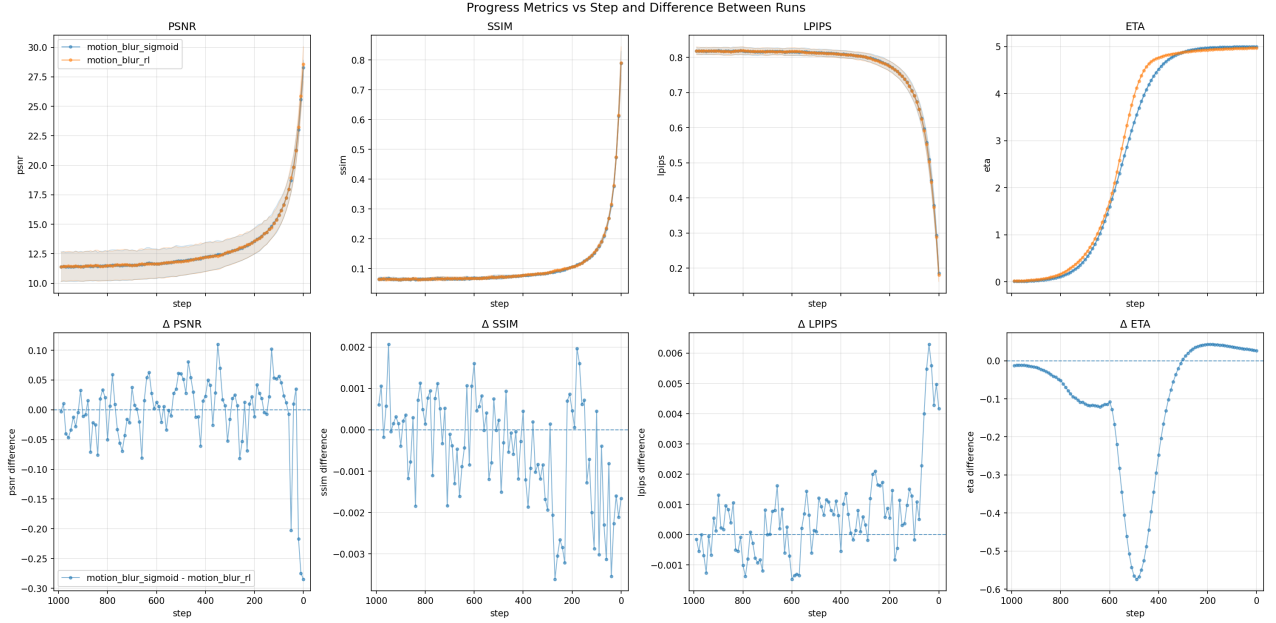


Figure 10: Comparison of sigmoid method with our RL method.

5.4 Comparison between the baseline and our RL model for different tasks

Table 5: Quantitative comparison of baseline and RL methods across different restoration tasks. Higher PSNR and SSIM are better, while lower LPIPS is better.

Task	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
motion blur	baseline	24.09 ± 2.12	0.71 ± 0.06	0.27 ± 0.04
	rl	28.58 ± 2.66	0.79 ± 0.06	0.18 ± 0.04
gaussian blur	baseline	25.94 ± 0.82	0.73 ± 0.05	0.23 ± 0.02
	rl	28.31 ± 0.82	0.77 ± 0.05	0.18 ± 0.01
inpainting random	baseline	30.49 ± 1.28	0.86 ± 0.03	0.17 ± 0.01
	rl	33.55 ± 1.46	0.88 ± 0.02	0.10 ± 0.01
inpainting box	baseline	23.94 ± 1.79	0.83 ± 0.03	0.19 ± 0.02
	rl	23.39 ± 1.73	0.82 ± 0.01	0.16 ± 0.03
super resolution	baseline	24.42 ± 0.76	0.72 ± 0.05	0.28 ± 0.02
	rl	27.76 ± 0.65	0.73 ± 0.04	0.27 ± 0.02

We can see that our RL model outperform the baseline model for every task and every metrics, in exception of the inpainting box task.

6 Expected Contributions of Each Team Member

- Kuan-Yu Lin: Reinforcement learning formulation and policy optimization. Responsible for defining the MDP, designing the reward function, and implementing the PPO-based policy learning framework.
- Joaquín Rus Bono: Diffusion model integration and inverse problem formulation. Responsible for implementing the diffusion-based reconstruction pipeline, including DPS and measurement-guided sampling baselines.
- Junjie Shao: Experimental evaluation and benchmarking. Responsible for dataset preparation, benchmark task implementation (inpainting, super-resolution, deblurring), and quantitative evaluation using PSNR, SSIM, and LPIPS.

The contributions described above are not absolute, as we are likely to help each other during each stage of the process.

References

References

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